

MD SABBIR HOSSEN

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🐙 GitHub ✦ 🌐 Personal Website ✦ 🎓 Google Scholar ✦ 🔗 LinkedIn

RESEARCH INTERESTS

Foundation and Multimodal AI

- Natural Language Processing (NLP)
- Large Language Models (LLMs)
- Computer Vision (CV)
- Vision-Language Models (VLMs)

Efficient and Resource-Constrained AI

- Efficient Deep Learning Architectures
- Low-Resource and Multilingual Language Processing
- Model Compression and Optimization
- AI for Low-Resource Settings

EDUCATION

Bangladesh University

B.Sc in Computer Science and Engineering

Mohammadpur, Dhaka-1207, Bangladesh

January 2020 - January 2024

CGPA: 3.66/4.00

Thesis: A Sophisticated Feature Vectorization-Based Stacked Machine Learning Approach for Fake News Detection in Bangla and English.

EXPERIENCES

Research Assistant

National Institute of Textile Engineering and Research

September 2025 - Present

- Collected, annotated, and preprocessed text data to build datasets for NLP research projects.
- Conducted literature reviews to identify research gaps and relevant methodologies.
- Developed and evaluated machine learning models and conducted research experiments.
- Collaborated and coordinated with faculty, researchers, and students from different universities and countries.

Research Assistant

Bangladesh University

May 2023 - July 2025

- Designed and implemented machine learning models for research projects in NLP and Computer Vision.
- Conducted literature reviews and wrote several research papers.
- Served as corresponding author for multiple research papers, handling reviewers' responses and revisions.
- Mentored three undergraduate research groups in research methodology and manuscript preparation.

SELECTED PUBLICATIONS

Md. Sabbir Hossen, Fahim Al Farid, Pabon Shaha, Md. Mowahibur Rahman Twake et al. "A Sophisticated Feature Vectorization-Based Stacked Machine Learning Approach for Fake News Detection in Bangla and English." *Social Network Analysis and Mining*, 2025. [Q1 Journal] **[Paper]**

Md. Sabbir Hossen, Md. Saiduzzaman, and Pabon Saha. "Social Media Sentiments Analysis on the July Revolution in Bangladesh: A Hybrid Transformer Based Machine Learning Approach." *In Proceedings of the IEEE 17th International Conference on Electronics, Computers and Artificial Intelligence (ECAI)*, 2025 **[Paper]**

Pabon Saha, **Md. Sabbir Hossen**, Md. Ibrahim Hosen Sojib, and Sanjida Akter et al. "StackTrace-AI: Identifying Generative AI Text Origins using Ensemble Learning." *In Proceedings of the IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025 **[Paper]**

Md. Sabbir Hossen and Md. Saiduzzaman. "TransCNN: A Hybrid CNN-Transformer Synergy for Reliable Deepfake Forensics." *In Proceedings of the IEEE 28th International Conference on Computer and Information Technology (ICCIT)*, 2025 **[Paper]**

Eshat Ahmed Shuvo, Md Shuvon, Nazmul Sarkar, and **Md. Sabbir Hossen** et al. "Optimized Hybrid Cascaded Approach for Accurate Oral Cancer Detection in Histopathology Images Using Deep CNNs." *In Proceedings of the IEEE 2nd International Conference on Next-Gen Computing, IoT and Machine Learning (NCIM)*, 2025 **[Paper]**

Md. Sabbir Hossen, Md. Saiduzzaman, Pabon Shaha, and Mostofa Kamal Nasir. "Jellyfish Species Identification: A CNN-Based Artificial Neural Network Approach." *In Proceedings of the IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN)*, 2025 **[Paper]**

Md. Sabbir Hossen, Pabon Saha, and Md. Saiduzzaman. "A Hybrid Machine Learning Approach Utilizing CNN Feature Extraction with Traditional Classifiers to Identify Strawberry Leaf Diseases." *In Proceedings of the IEEE 4th International Conference on Electrical, Computer and Communication Engineering (ECCE)*, 2025 [Paper]

Md. Sabbir Hossen, Pabon Shaha, and Md. Saiduzzaman et al. "An Explainable AI Driven Machine Learning Approach for Maternal Health Risk Analysis." *In Proceedings of the IEEE 27th International Conference on Computer and Information Technology (ICCIT)*, 2024 [Paper]

SELECTED SUBMITTED MANUSCRIPTS

Md. Tanvir Ahmed Shaikat, Pabon Shaha, Md. Sabbir Hossen, S M Rifatur Rana et al. "Automated Scoring of Bangla Student Responses: An Efficient and Explainable Deep Learning Approach," 2026 [Submitted to a peer-reviewed Journal]

Pabon Saha, Md. Sabbir Hossen, Mostofa Kamal Nasir, Anichur Rahman et al. "Catching the Bots: A Transformer-Based Ensemble Learning for Machine-Generated Text Detection," 2026 [Under Review in a peer-reviewed Journal]

Md. Sabbir Hossen, Eshat Ahmed Shuvo, Shibbir Ahmed Arif, Pabon Shaha, and Anichur Rahman et al. "An Efficient Deep Learning Framework for Brain Stroke Diagnosis Using Computed Tomography (CT) Images," 2025 [Under Review in a peer-reviewed Journal]

Eshat Ahmad Shuvo, Wahidur Rahman, Pabon Shaha, and Md. Sabbir Hossen et al. "Optimized Hybrid Approach for Early Detection of Alzheimer's Disease Using Machine Learning and Deep Learning Techniques," 2025 [Under Review in a peer-reviewed Journal]

CONFERENCE PRESENTATIONS

Conference Talk TransCNN: A Hybrid CNN-Transformer Synergy for Reliable Deepfake Forensics. *IEEE ICCIT*, 2025

Conference Talk Social Media Sentiments Analysis on the July Revolution in Bangladesh: A Hybrid Transformer Based Machine Learning Approach, *IEEE ECAI*, 2025

Conference Talk A Hybrid Machine Learning Approach Utilizing CNN Feature Extraction with Traditional Classifiers to Identify Strawberry Leaf Diseases, *IEEE QPAIN*, 2025

Conference Talk Jellyfish Species Identification: A CNN-Based Artificial Neural Network Approach, *IEEE ECCE*, 2025

Conference Talk An Explainable AI Driven Machine Learning Approach for Maternal Health Risk Analysis, *IEEE ICCIT*, 2024 [Best Technical Presentation]

ACADEMIC SERVICE

Journal Reviewer: Journal of Plant Diseases and Protection (JPDP), 2026

Conference Reviewer: IEEE USBEREIT, 2026

AWARDS & ACHIEVEMENTS

Best Technical Presentation, 27th International Conference on Computer and Information Technology, 2024

1st Runner Up in Project Showcasing, Robo Carnival, BUET, 2023

PROFESSIONAL SKILLS

Programming Languages: Python, C, C++

Python Libraries: NumPy, Pandas, scikit-learn, Matplotlib

ML Frameworks: PyTorch, TensorFlow, Hugging Face Transformers

Software Tools: Git, GitHub, Microsoft Office, LaTeX, Mendeley

EXTRA-CURRICULAR ACTIVITIES

iTech Expo associated with Techfest IIT Bombay, IUBAT November 2022
Presented a project on Remote Control Fire Fighter Robot.

Robo Carnival, BUET January 2023
Presented a project on Green Energy E-Bike.

BEC Robo Mania, BUBT January 2023
Presented a project on an Integrated Smart Home.

TEST SCORE

IELTS 7.0